3. **Interactivity**: Provides a range of widgets such as sliders, text inputs, and buttons to make applications interactive.
4. **Visualization**: Easily integrates with popular Python visualization libraries like Matplotlib, Plotly, and Seaborn to display data and model predictions.
5. **Open Source and Free**: No licensing costs, making it an ideal choice for hobbyists, researchers, and professionals.

Setting Up a Streamlit Application

Below is an example of deploying a machine learning model using Streamlit.

Example: Deploying a Model with Streamlit

```
import streamlit as st
import joblib

# Load the model
model = joblib.load('linear_regression_model.pkl') # Create a Streamlit app
st.title("AI Model Deployment with Streamlit") # Input features
feature = st.number_input("Enter a feature value:") # Predict button
if st.button("Predict"):
    prediction = model.predict([[feature]])[0]
    st.write(f"Prediction: {prediction}")
```

Step-by-Step Guide to Deploy with Streamlit

1. **Install Streamlit**: Install Streamlit using pip:

```
pip install streamlit
```

2. **Write the Application Script**: Create a Python script (e.g., app.py) that includes Streamlit components such as:
    - **st.title**: For adding a title to your application.
    - **st.input and st.number_input**: For capturing user input.
    - **st.write**: For displaying output and results.
3. **Run the Application**: Launch the Streamlit app using the following command:

```
streamlit run app.py
```

This will start a local web server and open the application in your default browser.